Integer Linear Programming
Algorithmic Methods for Mathematical Models

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Problem 1: Food Production Optimization

Problem Statement

A food is manufactured by refining raw oils and blending them together. The raw oils come in two categories:
Vegetable oil: VEG 1, VEG 2
Non-vegetable oil: NON 1, NON 2, NON 3
Vegetable oils and non-vegetable oils require different production lines for refining. It is not possible to refine more than 200 tons of vegetable oil and more than 250 tons of non-vegetable oils. There is no loss of weight in the refining process and the cost of refining may be ignored.
There is a technological restriction of hardness in the final product. In the units in which hardness is measured, this must lie between 3 and 6. We assume the hardness of final product is the weighted average of hardness of the raw oils.
Data:

	VEG 1	VEG 2	NON 1	NON 2	NON 3
Hardness	8.8	6.1	2.0	4.2	5.0
Cost (€/ton)	110	120	130	110	115

The final product sells at 150€ per ton.
Additional conditions:

- The food should be made up of at least three oils.
- If an oil is used then at least 20 tons must be used.
- If either of VEG 1 or VEG 2 is used then NON 3 must also be used.

Objective: What manufacturing policy should the company pursue in order to maximize profit? Formulate the problem as an MIP.

Part (a): Basic LP Formulation

Decision variables:

$$TVEG_1, TVEG_2 = \text{tons of vegetable oils VEG1, VEG2 used} \tag{1}$$
$$TNON_1, TNON_2, TNON_3 = \text{tons of non-vegetable oils NON1, NON2, NON3 used} \tag{2}$$
$$PROD = TVEG_1 + TVEG_2 + TNON_1 + TNON_2 + TNON_3 \text{ (total production)} \tag{3}$$

Objective function:

$$\text{Maximize } Z = 150 \cdot PROD - 110 \cdot TVEG_1 - 120 \cdot TVEG_2 \tag{4}$$
$$- 130 \cdot TNON_1 - 110 \cdot TNON_2 - 115 \cdot TNON_3 \tag{5}$$

Basic constraints:

$$\text{Vegetable oil refining capacity: } TVEG_1 + TVEG_2 \leq 200 \tag{6}$$

$$\text{Non-vegetable oil refining capacity: } TNON_1 + TNON_2 + TNON_3 \leq 250 \tag{7}$$

$$\text{Minimum hardness:} \tag{8}$$

$$\frac{8.8 \cdot TVEG_1 + 6.1 \cdot TVEG_2 + 2.0 \cdot TNON_1 + 4.2 \cdot TNON_2 + 5.0 \cdot TNON_3}{PROD} \geq 3 \tag{9}$$

$$\text{Maximum hardness:} \tag{10}$$

$$\frac{8.8 \cdot TVEG_1 + 6.1 \cdot TVEG_2 + 2.0 \cdot TNON_1 + 4.2 \cdot TNON_2 + 5.0 \cdot TNON_3}{PROD} \leq 6 \tag{11}$$

$$\text{Total production definition:} \tag{12}$$

$$PROD = TVEG_1 + TVEG_2 + TNON_1 + TNON_2 + TNON_3 \tag{13}$$

$$TVEG_1, TVEG_2, TNON_1, TNON_2, TNON_3 \geq 0 \tag{14}$$

Part (b): MIP Formulation with Additional Conditions

Additional binary variables:

$$UVEG_1, UVEG_2 = \begin{cases} 1 & \text{if vegetable oil VEG1, VEG2 is used} \\ 0 & \text{otherwise} \end{cases} \tag{15}$$

$$UNON_1, UNON_2, UNON_3 = \begin{cases} 1 & \text{if non-vegetable oil NON1, NON2, NON3 is used} \\ 0 & \text{otherwise} \end{cases} \tag{16}$$

Additional constraints for extra conditions:

$$\text{At least 3 oils: } UVEG_1 + UVEG_2 + UNON_1 + UNON_2 + UNON_3 \geq 3 \tag{17}$$

$$\text{Minimum usage per oil:} \tag{18}$$

$$TVEG_1 \geq 20 \cdot UVEG_1, \quad TVEG_2 \geq 20 \cdot UVEG_2 \tag{19}$$

$$TNON_1 \geq 20 \cdot UNON_1, \quad TNON_2 \geq 20 \cdot UNON_2, \quad TNON_3 \geq 20 \cdot UNON_3 \tag{20}$$

$$\text{Maximum usage per oil:} \tag{21}$$

$$TVEG_1 \leq M \cdot UVEG_1, \quad TVEG_2 \leq M \cdot UVEG_2 \tag{22}$$

$$TNON_1 \leq M \cdot UNON_1, \quad TNON_2 \leq M \cdot UNON_2, \quad TNON_3 \leq M \cdot UNON_3 \tag{23}$$

$$\text{VEG}\rightarrow\text{NON3 implication: } UVEG_1 + UVEG_2 \leq 2 \cdot UNON_3 \tag{24}$$

$$UVEG_1, UVEG_2, UNON_1, UNON_2, UNON_3 \in \{0, 1\} \tag{25}$$

Binary Variable Transformations

How to find the transformations for binary variables?
Let b be a binary variable and $f(x)$ a linear expression.
Case 1: We want to express with linear inequalities that "if $b = 0$ then $f(x) \leq 0$":
Let U be an upper bound for the value of $f(x)$ over all feasible solutions. Then, for any feasible solution, the following constraint holds:

$$f(x) \leq U \cdot b$$

If $b = 0$, then $f(x) \leq 0$. If $b = 1$, then $f(x) \leq U$, which is always true.
Case 2: We want to express with linear inequalities that "if $b = 1$ then $f(x) \geq 0$":
Let L be a lower bound for the value of $f(x)$ over all feasible solutions. Then, for any feasible solution, the following constraint holds:

$$f(x) \geq L \cdot (1 - b)$$

If $b = 1$, then $f(x) \geq 0$. If $b = 0$, then $f(x) \geq L$, which is always true.

In case we have "if $b = 0$ then $f(x) \geq k$": We can transform our function by defining a new function $f'(x) = f(x) - k$. Then, we can apply the same logic as in Case 2: Let L' be a lower bound for the value of $f'(x)$ over all feasible solutions. Then, for any feasible solution, the following constraint holds:

$$f'(x) \geq L' \cdot (1 - b)$$

Which can be rewritten as:

$$f(x) - k \geq L' \cdot (1 - b)$$

or equivalently:

$$f(x) \geq k + L' \cdot (1 - b)$$

Problem 2: Sudoku

Given the following Sudoku puzzle,

5	3			7				
6			1	9	5			
	9	8					6	
8				6				3
4			8		3			1
7				2				6
	6					2	8	
			4	1	9			5
				8			7	9

formulate the problem of completing it as a MIP.

MIP Formulation

Decision variables:

- x_{ijk}: binary variable that is 1 if the cell in row i and column j contains the number k, and 0 otherwise. Where $i, j, k \in \{1, 2, \ldots, 9\}$.

Objective function: There is no objective function. To cost the problem as a MIP, we can take arbitrarily any function, for example: we can minimize 0.

Constraints:

1. At each row, each value appears exactly once:

$$\sum_{i=1}^{9} x_{ijk} = 1 \quad \forall j, k \in \{1, \ldots, 9\}$$

2. At each column, each value appears exactly once:

$$\sum_{j=1}^{9} x_{ijk} = 1 \quad \forall i, k \in \{1, \ldots, 9\}$$

3. In each 3x3 subsquare, each value appears exactly once:

   Let's us identify each of the subsquares with the set of coordinates if its cells:

   - $\mathbf{S_{11}} = \{(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)\}$

$$\begin{bmatrix} S_{11} & S_{12} & S_{13} \\ S_{21} & S_{22} & S_{23} \\ S_{31} & S_{32} & S_{33} \end{bmatrix}$$

   For each subsquare S, for all values k:

$$\sum_{(i,j) \in S} x_{ijk} = 1 \quad \forall k \in \{1, \ldots, 9\}$$

4. Fixed cells are respected.
   For each fixed cell (i,j) with value k:

$$x_{ijk} = 1$$

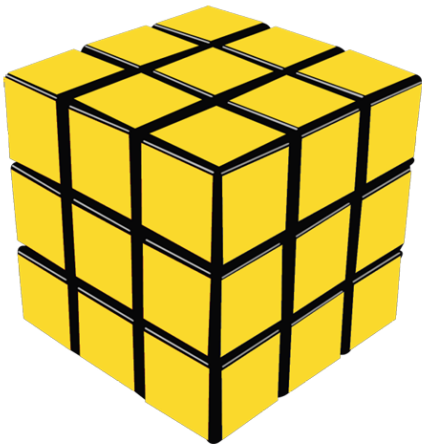
5. At each cell (i,j), there is exactly one value:

$$\sum_{k=1}^{9} x_{ijk} = 1 \quad \forall i, j \in \{1, \ldots, 9\}$$

Problem 3: Twenty-Seven Cells Arrangement

Problem Statement

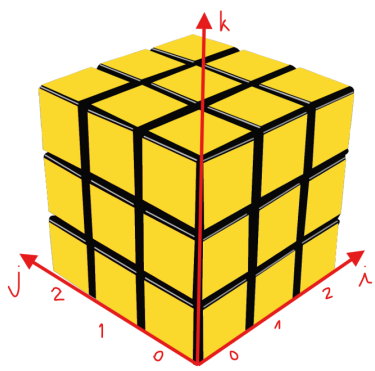
Twenty-seven cells are arranged in a three-dimensional $3 \times 3 \times 3$ cube as shown below.



Three cells are regarded as lying in the same line if they are on the same horizontal or vertical line or the same diagonal. Diagonals exist on each horizontal and vertical section and connecting opposite vertices of the cube. (There are 49 lines altogether.)
Given 13 white balls and 14 black balls, arrange them, one to a cell, so as to minimise the number of lines with balls all of one colour.

MIP Formulation

Let's identify each of the three cells in the cube with a triplet (i, j, k) where $i, j, k \in \{0, 1, 2\}$.
Moreover, let's identify a line with the set of coordinates of its cells.



Decision variables:

- x_{ijk} = the ball at cell (i, j, k) is white (1) or black (0).
  Where $i, j, k \in \{0, 1, 2\}$. (of type binary)
- y_l = the ball in line l are all of the same color (1) or not (0).
  Where $l \in \{1, 2, \ldots, 49\}$ (the 49 lines in total). (of type binary)

Objective function: Minimize the number of lines with balls all of one colour:

$$\text{minimize} \sum_{l=1}^{49} y_l$$

Constraints:

1. There are 13 white balls and 14 black balls:

$$\boxed{\sum_{(i,j,k)\in\text{cells}} x_{ijk} = 13} \quad \text{and} \quad \boxed{\sum_{(i,j,k)\in\text{cells}} (1 - x_{ijk}) = 14} \tag{26}$$

2. If all balls in a line l have the same colour, then $y_l = 1$.
   Let l be the line consisting of the cells $(i_0, j_0, k_0), (i_1, j_1, k_1), (i_2, j_2, k_2)$. Let's consider two cases:

   (a) If all balls in line l are **white** , then $y_l = 1$:

$$x_{i_0 j_0 k_0} \wedge x_{i_1 j_1 k_1} \wedge x_{i_2 j_2 k_2} \Rightarrow y_l$$

$$\neg(x_{i_0 j_0 k_0} \wedge x_{i_1 j_1 k_1} \wedge x_{i_2 j_2 k_2}) \vee y_l \quad \text{(by implication law)}$$

   If $a_1, a_2, \ldots, a_n, b_1, b_2, \ldots, b_m$ are binary variables, then:

$$\neg(a_1 \wedge a_2 \wedge \cdots \wedge a_n) \vee (b_1 \wedge b_2 \wedge \cdots \wedge b_m) \Leftrightarrow \sum_{i=1}^{n}(1 - a_i) + \sum_{j=1}^{m} b_j \geq 1$$

   Therefore, we can express our constraint as:

$$\boxed{(1 - x_{i_0 j_0 k_0}) + (1 - x_{i_1 j_1 k_1}) + (1 - x_{i_2 j_2 k_2}) + y_l \geq 1}$$

   (b) If all balls in line l are **black** , then $y_l = 1$:

$$(1 - x_{i_0 j_0 k_0}) \wedge (1 - x_{i_1 j_1 k_1}) \wedge (1 - x_{i_2 j_2 k_2}) \Rightarrow y_l$$

$$\neg((1 - x_{i_0 j_0 k_0}) \wedge (1 - x_{i_1 j_1 k_1}) \wedge (1 - x_{i_2 j_2 k_2})) \vee y_l$$

   Applying the same reasoning as in the previous case, we can express our constraint as:

$$\boxed{x_{i_0 j_0 k_0} + x_{i_1 j_1 k_1} + x_{i_2 j_2 k_2} + y_l \geq 1}$$

Problem 4: Infectious Disease Outbreak Investigation

Problem Statement

An outbreak of an infectious disease has been observed in a set N of locations. There exists a set M of teams capable of investigating these outbreaks. Team $i \in M$ can conclude its investigation of the outbreak in location $j \in N$ in t_{ij} hours. Each team can either investigate zero, one, or two outbreaks. If a team investigates two outbreaks, they must travel from one location to the next. The travel time from location $j_1 \in N$ to location $j_2 \in N$ is $d_{j_1 j_2}$. Travel times can be assumed to be symmetric. Once all outbreaks have been investigated, a disease control center can take action to combat the outbreak. The goal is to minimize the amount of time necessary to complete the investigation of all locations.

MIP Formulation

Decision variables:

- $x_{ij} = 1$ if team i investigates location j, 0 otherwise.
  Where $1 \leq i \leq M$ and $1 \leq j \leq N$. (of type binary)
- $y_{ij_1j_2} = 1$ if team i investigates locations j_1 and j_2 in that order, 0 otherwise.
  Where $1 \leq i \leq M$ and $1 \leq j_1, j_2 \leq N$. (of type binary)
- z = the maximum time needed for any team to complete its investigations. (of type real non negative)

Constraints:

1. Each team can investigate at most two locations:

$$\boxed{\sum_{j=1}^{N} x_{ij} \leq 2 \quad \forall 1 \leq i \leq M}$$

2. Each location must be investigated by exactly one team:

$$\boxed{\sum_{i=1}^{M} x_{ij} = 1 \quad \forall 1 \leq j \leq N}$$

3. Variables are linked: $y_{ij_1j_2} \iff (x_{ij_1} \wedge x_{ij_2})$:

   (a) $y_{ij_1j_2} \implies (x_{ij_1} \wedge x_{ij_2})$:

$$\neg y_{ij_1j_2} \vee (x_{ij_1} \wedge x_{ij_2})$$
$$(\neg y_{ij_1j_2} \vee x_{ij_1}) \wedge (\neg y_{ij_1j_2} \vee x_{ij_2})$$
$$(1 - y_{ij_1j_2}) + x_{ij_1} \geq 1 \text{ and } (1 - y_{ij_1j_2}) + x_{ij_2} \geq 1$$

   Therefore, we can express our constraints as:

$$\boxed{y_{ij_1j_2} \leq x_{ij_1}} \quad \text{and} \quad \boxed{y_{ij_1j_2} \leq x_{ij_2}} \quad \forall 1 \leq i \leq M, 1 \leq j_1, j_2 \leq N$$

   (b) $y_{ij_1j_2} \impliedby (x_{ij_1} \wedge x_{ij_2})$:

$$(x_{ij_1} \wedge x_{ij_2}) \implies y_{ij_1j_2}$$
$$\neg(x_{ij_1} \wedge x_{ij_2}) \vee y_{ij_1j_2}$$
$$\neg x_{ij_1} \vee \neg x_{ij_2} \vee y_{ij_1j_2}$$
$$(1 - x_{ij_1}) + (1 - x_{ij_2}) + y_{ij_1j_2} \geq 1$$

   Therefore, we can express our constraint as:

$$\boxed{x_{ij_1} + x_{ij_2} - y_{ij_1j_2} \leq 1} \quad \forall 1 \leq i \leq M, 1 \leq j_1, j_2 \leq N$$

4. z has its intended meaning: For every team i, the time needed for team i to complete its investigations must be less than or equal to z:

$$\boxed{\sum_{j=1}^{N} t_{ij} x_{ij} + \sum_{j_1, j_2 = 1}^{N} \sum_{j_1 < j_2} d_{j_1 j_2} \cdot y_{ij_1j_2} \leq z \quad \forall 1 \leq i \leq M}$$

Objective function: Minimize the time needed to complete the investigation of all locations.

$$\min \max_{1 \leq i \leq M} \{\text{time needed for team } i\}$$

The time needed for team i to complete its investigations is

$$\sum_{j=1}^{N} t_{ij} x_{ij} + \sum_{j_1=1}^{N} \sum_{j_2=1}^{N} d_{j_1 j_2} (x_{ij_1} \wedge x_{ij_2}) \implies \sum_{\substack{j_1, j_2 = 1 \\ j_1 < j_2}}^{N} d_{j_1 j_2} \cdot y_{ij_1 j_2}$$

Problem 5: Graph Coloring

Problem Statement

Consider the problem of graph coloring: given a graph $G = (V, E)$, we want to find the minimum number of colors to paint the vertices so that adjacent vertices have different colors. Model this problem as a MIP.

MIP Formulation

Let's label vertices $1, 2, \ldots, n = |V|$.

Decision variables:

- c_v = color assigned to vertex v (for all $v \in V$) (of type integer non negative)
- z = intuitively, $\max_{v \in V} c_v$ (of type integer non negative)
- $z_{uv} = 1$ if $c_u < c_v$, 0 if $c_u > c_v$ (for all $(u, v) \in E$) (of type binary)

Objective function: Minimize the number of colors used:

$$\min z$$

Constraints:

1. z 'behaves' as $\max_{v \in V} c_v$:

$$\boxed{c_v \leq z \quad \forall v \in V}$$

2. Adjacent vertices have different colors:

$$\boxed{c_u \neq c_v \quad \forall (u, v) \in E}$$

The problem is that this equation is not linear. We can transform it using binary variables:
This is the same as saying that either $c_u < c_v$ or $c_u > c_v \implies c_u + 1 \leq c_v \vee c_u \geq c_v + 1$
We define a new variable z_{uv} such that:

$$z_{uv} = \begin{cases} 1 & \text{if } c_u < c_v \\ 0 & \text{if } c_u > c_v \end{cases}$$

We can express our constraint as:

$$z_{uv} = 1 \implies c_u + 1 \leq c_v \implies c_v - c_u - 1 \geq 0$$

$$z_{uv} = 0 \implies c_u \geq c_v + 1 \implies c_u - c_v - 1 \geq 0$$

Applying the transformations for binary variables explained in Problem 1, we get:
If $b = 0 \longrightarrow f(x) \leq 0$ is equivalent to $f(x) \leq U \cdot b$, where U is an upper bound for $f(x)$.
If $b = 1 \longrightarrow f(x) \geq 0$ is equivalent to $f(x) \geq L \cdot (1 - b)$, where L is a lower bound for $f(x)$.

- Let's express $z_{uv} = 1 \implies c_v - c_u - 1 \geq 0$ We know that $c_v \geq 1 \implies c_u \leq n \implies -c_u \geq -n$
  Therefore, a lower bound is $L = -n$.
  So, the implication is equivalent to:

$$c_v - c_u - 1 \geq -n \cdot (1 - z_{uv})$$

- Let's express $z_{uv} = 0 \implies c_u - c_v - 1 \geq 0$ We know that $c_v \leq n \implies c_u \geq 1 \implies -c_u \leq -1$
  Therefore, an upper bound is $U = n$.
  So, the implication is equivalent to:

$$c_u - c_v - 1 \leq n \cdot z_{uv}$$

Therefore, we can express our constraints as:

$$\boxed{c_v - c_u - 1 \geq -n \cdot (1 - z_{uv})} \quad \text{and} \quad \boxed{c_u - c_v - 1 \leq n \cdot z_{uv}} \quad \forall (u, v) \in E$$

Alternative MIP Formulation

Another possibility of defining the decision variables is:

- $x_{vc} = 1$ if vertex v is assigned color c, 0 otherwise (for all $v \in V$ and $1 \leq c \leq n$) (of type binary)
- $u_k = 1$ if color k is used, 0 otherwise (for all $1 \leq k \leq n$) (of type binary)
- $z =$ number of colors used (of type integer non negative)

Objective function: Minimize the number of colors used:

$$\min z$$

Constraints:

1. Variables are related: For each $1 \leq v \leq n$ and $1 \leq k \leq n$:

$$x_{vk} \to u_k \quad \Longrightarrow \quad \neg x_{vk} \vee u_k \quad \Longrightarrow \quad (1 - x_{vk}) + u_k \geq 1 \quad \Longrightarrow \quad \boxed{u_k \geq x_{vk}}$$

2. Neighbors are painted with different colors: For each edge $(u, v) \in E$ and $1 \leq k \leq n$:

$$\neg(x_{uk} \wedge x_{vk}) \quad \Longrightarrow \quad (1 - x_{uk}) + (1 - x_{vk}) \geq 1 \quad \Longrightarrow \quad \boxed{x_{uk} + x_{vk} \leq 1}$$

3. Each vertex is painted with exactly one color: For each $1 \leq v \leq n$:

$$\boxed{\sum_{k=1}^{n} x_{vk} = 1}$$

Observation: This constraint was not in our first formulation because it was implicit in the definition of the variables c_v.

Problem 6: Medical Drug Acquisition Planning

Problem Statement

A medical company is attempting to acquire a certain drug from a set M of suppliers. The company requires d_t units of this drug in month t, for $t = 1, \ldots, T$, which are entirely consumed. Purchasing one unit of the drug from supplier $i \in M$ during time period $t \in \{1, \ldots, T\}$ costs c_{it} (in €). However, in order to purchase drugs from supplier $i \in M$, at any time period, the company must purchase a minimum of ℓ_i units during that time period. Fortunately, if it turns out that there is more drug than required, the company can store at most h units from one period to the next. If even then the company finds itself with too many units of the drug, it can simply throw away the extra supply. The goal is to minimize the cost required to purchase the drug for time periods $1, \ldots, T$.

MIP Formulation

Decision variables:

- x_{ijt} = number of units of drug purchased from supplier i at time period t (for all $1 \leq i \leq M$ and $1 \leq t \leq T$) (of type integer non negative)
- y_t = number of units of drug stored at the end of time period t (for all $1 \leq t \leq T-1$) (of type integer non negative) (for convinience, we define $y_0 = 0$)
- $z_{it} = 1$ if we purchase drugs from supplier i at time period t, 0 otherwise (for all $1 \leq i \leq M$ and $1 \leq t \leq T$) (of type binary)

Objective function: Minimize costs:

$$\min \sum_{\substack{1 \leq i \leq M \\ 1 \leq t \leq T}} c_{it} \cdot x_{it}$$

Constraints:

1. At every period t, the demand d_t must be satisfied:

$$\begin{aligned}
&\text{For period } t = 1: && \sum_{i=1}^{M} x_{i1} \geq d_1 \\
&\text{For period } t = 2: && \sum_{i=1}^{M} x_{i2} + y_1 \geq d_2 + y_2 \\
&\vdots \\
&\text{For period } t = T: && \sum_{i=1}^{M} x_{iT} + y_{T-1} \geq d_T + y_T
\end{aligned}$$

   We can express our constraints as:

$$\boxed{\sum_{i=1}^{M} x_{it} + y_{t-1} \geq d_t + y_t \quad \forall 1 \leq t \leq T}$$

   Explanation of the constraint: the total amount of drug available at period t (the amount purchased at period t plus the amount stored from the previous period) must be greater than or equal to the total amount of drug needed at period t (the amount required at period t plus the amount stored for the next period).

2. The storage capacity is limited:

$$\boxed{y_t \leq h \quad \forall 1 \leq t \leq T}$$

3. The purchase of drugs respect the minimum order quantity: For all $1 \leq i \leq M$ and $1 \leq t \leq T$:

   'If we buy drugs from supplier i at time period t, then we must buy at least ℓ_i units' can be expressed as:

$$z_{it} = 0 \rightarrow x_{it} \leq 0 \quad \text{and} \quad z_{it} = 1 \rightarrow x_{it} \geq \ell_i$$

   Applying the transformations for binary variables, we get:

$$\boxed{x_{it} \leq U_{iz} \cdot z_{it}} \quad \text{and} \quad \boxed{x_{it} \geq \ell_i \cdot z_{it}} \quad \forall 1 \leq i \leq M, 1 \leq t \leq T$$

   Where U_{iz} is an upper bound for the value of x_{it} over all feasible solutions. A possible value for U_{iz} is $\sum_{t=1}^{T} d_t$.

Problem 7: Nurse Scheduling Problem

Problem Statement

A nurse is assigned a set N of patients. For each patient $i \in N$, the nurse must spend p_i minutes examining the patient. The nurse must then wait somewhere between ℓ_i and u_i minutes before checking up on the patient, which requires q_i minutes. The nurse, of course, cannot be in two places at the same time, although we will assume for simplicity that the travel time to walk from one patient's room to another is zero. The objective is to minimize the total amount of time required to tend to all patients.

MIP Formulation

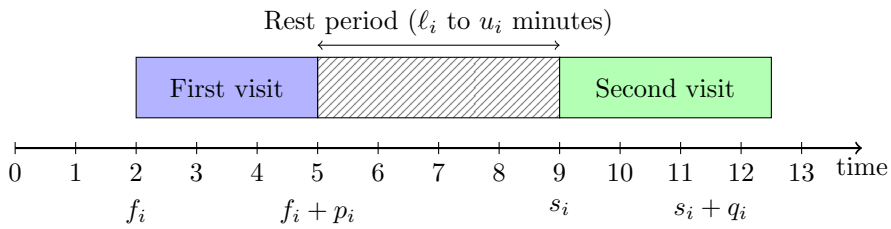
Decision variables:

- f_i = time the first visit to patient i starts (for all $i \in N$) (of type real non negative)
- s_i = time the second visit to patient i starts (for all $i \in N$) (of type real non negative)
- ff_{ij} = 1st visit to patient i is before 1st visit to patient j, (for all $i, j \in N, i < j$) (of type binary)
- fs_{ij} = 1st visit to patient i is before 2nd visit to patient j, (for all $i, j \in N$) (of type binary)
- sf_{ij} = 2nd visit to patient i is before 1st visit to patient j, (for all $i, j \in N$) (of type binary)
- ss_{ij} = 2nd visit to patient i is before 2nd visit to patient j, (for all $i, j \in N, i < j$) (of type binary)

Constraints:

1. The rest between visits is respected:

$$\boxed{\ell_i \leq s_i - (f_i + p_i) \leq u_i \quad \forall i \in N}$$



Explanation:

- f_i = Time when first visit to patient i starts
- p_i = Duration of first visit (examination)
- $f_i + p_i$ = Time when first visit ends
- s_i = Time when second visit (check-up) starts
- $s_i - (f_i + p_i)$ = Rest time between visits
- This rest time must be between ℓ_i and u_i minutes

2. There are no overlapping visits:

   For each pair of patients $i, j \in N$ with $i < j$, we have four possible cases:

   (a) First visit to patient i ends before first visit to patient j starts:
   (b) First visit to patient j ends before first visit to patient i starts:
   (c) Second visit to patient i ends before second visit to patient j starts:
   (d) Second visit to patient j ends before second visit to patient i starts:

   We will define four binary variables to represent these cases. See in the decision variables section.

   Therefore, we can express our constraints as:

$$ff_{ij} + ff_{ji} = 1 \quad \forall i, j \in N, i < j$$
$$ss_{ij} + ss_{ji} = 1 \quad \forall i, j \in N, i < j$$
$$fs_{ij} + sf_{ji} = 1 \quad \forall i, j \in N, i \neq j$$

3. The binary variables are linked to the time variables:

   (a) $ff_{ij} = 1 \rightarrow f_i + p_i \leq f_j$:
   $$f_i + p_i \leq f_j + U \cdot (1 - ff_{ij})$$
   Where U is an upper bound for the value of $f_j - (f_i + p_i)$ over all feasible solutions. A possible value for U is $\sum_{i \in N}(p_i + u_i + q_i)$.

   (b) $ss_{ij} = 1 \rightarrow s_i + q_i \leq s_j$:
   $$s_i + q_i \leq s_j + U \cdot (1 - ss_{ij})$$
   Where U is an upper bound for the value of $s_j - (s_i + q_i)$ over all feasible solutions. A possible value for U is $\sum_{i \in N}(p_i + u_i + q_i)$.

   (c) $fs_{ij} = 1 \rightarrow f_i + p_i \leq s_j$:
   $$f_i + p_i \leq s_j + U \cdot (1 - fs_{ij})$$
   Where U is an upper bound for the value of $s_j - (f_i + p_i)$ over all feasible solutions. A possible value for U is $\sum_{i \in N}(p_i + u_i + q_i)$.

   (d) $sf_{ij} = 1 \rightarrow s_i + q_i \leq f_j$:
   $$s_i + q_i \leq f_j + U \cdot (1 - sf_{ij})$$
   Where U is an upper bound for the value of $f_j - (s_i + q_i)$ over all feasible solutions. A possible value for U is $\sum_{i \in N}(p_i + u_i + q_i)$.

Objective function:
Maximize the total profit:
$$\max \sum_{m \in M} \sum_{i \in P} \text{Profit}_i \cdot x_{i,m}$$

Problem 8. Factories

A company has two factories, one at Liverpool and one at Brighton. In addition, it has four depots with storage facilities at Newcastle, Birmingham, London and Exeter. The company sells its product to six customers C1, C2, ..., C6. Customers can be supplied from a or the factory directly.
The distribution costs are known; they are given in the table below (in pounds per ton delivered). A dash in the table indicates the impossibility of certain suppliers for certain depots or customers.

	Liverpool	Brighton	Newcastle	Birmingham	London	Exeter
Newcastle	0.5	-				
Birmingham	0.5	0.3				
London	1.0	0.5				
Exeter	0.2	0.2				
C1	1.0	2.0	-	1.0	-	-
C2	-	-	1.5	0.5	1.5	-
C3	1.5	-	0.5	0.5	2.0	0.2
C4	2.0	-	1.5	1.0	-	1.5
C5	-	-	-	0.5	0.5	0.5
C6	1.0	-	1.0	-	1.5	1.5

Each factory has a monthly capacity given as follows, which cannot be exceeded:

Factory	Monthly Capacity
Liverpool	150,000 tons
Brighton	200,000 tons

Table 1: Monthly Capacity for Each Factory

Each depot has a maximum monthly throughput given as follows, which cannot be exceeded:

Depot	Maximum Monthly Throughput
Newcastle	70,000 tons
Birmingham	50,000 tons
London	100,000 tons
Exeter	40,000 tons

Table 2: Maximum Monthly Throughput for Each Depot

Each customer has a monthly requirement given as follows, which must be met:

Customer	Monthly Requirement
C1	50,000 tons
C2	10,000 tons
C3	40,000 tons
C4	35,000 tons
C5	60,000 tons
C6	20,000 tons

Table 3: Monthly Requirements for Each Customer

(a) The company would like to determine what distribution pattern minimises overall cost. Formulate the problem as an LP.
(b) There is a possibility of opening new depots at Bristol and Northampton, as well as of enlarging the Birmingham depot. It is not considered desirable to have more than four depots, and if necessary Newcastle or Exeter (or both) can be closed down. The monthly costs (in interest charges) of the possible new depots and expansion at Birmingham are given in the table below together with the potential monthly throughputs.

Depot	Monthly Cost (1000 pounds)	Monthly Throughput (1000 tons)
Bristol	12	30
Northampton	4	25
Birmingham (expansion)	3	20

Table 4: Monthly Costs and Throughput for New Depots and Expansion

The monthly savings of closing down the Newcastle and Exeter depots are given in the following table:

Depot	Monthly Savings (1000 pounds)
Newcastle	10
Exeter	5

Table 5: Monthly Savings from Closing Depots

The distribution costs involving the new depots are given in the following tables (in pounds per ton delivered)

Supplied to	Supplier	
	Liverpool	Brighton
New depots		
Bristol	0.6	0.4
Northampton	0.4	0.3

Supplied to	Supplier	
	Bristol	Northampton
Customers		
C1	1.2	
C2	0.6	0.4
C3	0.5	
C4		0.5
C5	0.3	0.6
C6	0.8	0.9

What would be the best resultant distribution pattern to minimise overall costs? Formulate the problem as a MIP. Could you formulate it as an LP?
Solution:

(a) LP Formulation

Decision variables:

- $x_{f,d}$ = amount of product shipped from factory f to depot d (for all factories f and depots d) (of type real non negative)
- $y_{d,c}$ = amount of product shipped from depot d to customer c (for all depots d and customers c) (of type real non negative)
- $z_{f,c}$ = amount of product shipped directly from factory f to customer c (for all factories f and customers c) (of type real non negative)

Sets:

- Factories: $F = \{\text{Liverpool}, \text{Brighton}\}$
- Depots: $D = \{\text{Newcastle}, \text{Birmingham}, \text{London}, \text{Exeter}\}$
- Customers: $C = \{\text{C1}, \text{C2}, \text{C3}, \text{C4}, \text{C5}, \text{C6}\}$

Objective function: Minimize total distribution cost:

$$\min \sum_{f \in F} \sum_{d \in D} c_{f,d} \cdot x_{f,d} + \sum_{d \in D} \sum_{c \in C} c_{d,c} \cdot y_{d,c} + \sum_{f \in F} \sum_{c \in C} c_{f,c} \cdot z_{f,c}$$

Constraints:

1. Factory capacity constraints:

$$\boxed{\sum_{d \in D} x_{f,d} + \sum_{c \in C} z_{f,c} \leq \text{Capacity}_f \quad \forall f \in F}$$

2. Depot throughput constraints:

$$\boxed{\sum_{f \in F} x_{f,d} \geq \sum_{c \in C} y_{d,c} \quad \forall d \in D}$$

$$\boxed{\sum_{f \in F} x_{f,d} \leq \text{MaxThroughput}_d \quad \forall d \in D}$$

3. Customer demand constraints:

$$\boxed{\sum_{d \in D} y_{d,c} + \sum_{f \in F} z_{f,c} = \text{Demand}_c \quad \forall c \in C}$$

(b) MIP Formulation Decision variables:

- $x_{f,d}$ = amount of product shipped from factory f to depot d (for all factories f and depots d) (of type real non negative)
- $y_{d,c}$ = amount of product shipped from depot d to customer c (for all depots d and customers c) (of type real non negative)
- $z_{f,c}$ = amount of product shipped directly from factory f to customer c (for all factories f and customers c) (of type real non negative)
- $w_d = 1$ if depot d is open, 0 otherwise (for all depots d) (of type binary)
- $e_b = 1$ if Birmingham depot is expanded, 0 otherwise (of type binary)

Objective function: Minimize total distribution cost including depot costs and savings from closures:

$$\min \sum_{f \in F} \sum_{d \in D} c_{f,d} \cdot x_{f,d} + \sum_{d \in D} \sum_{c \in C} c_{d,c} \cdot y_{d,c} + \sum_{f \in F} \sum_{c \in C} c_{f,c} \cdot z_{f,c} + \sum_{d \in D_{\text{new}}} \text{Cost}_d \cdot w_d + \text{Cost}_{\text{Birmingham}} \cdot e_b - \sum_{d \in D_{\text{close}}} \text{Savings}_d \cdot (1 - w_d)$$

Constraints:

1. Factory capacity constraints:

$$\boxed{\sum_{d \in D} x_{f,d} + \sum_{c \in C} z_{f,c} \leq \text{Capacity}_f \quad \forall f \in F}$$

2. Depot throughput constraints with opening decision:

$$\boxed{\sum_{f \in F} x_{f,d} \geq \sum_{c \in C} y_{d,c} \quad \forall d \in D}$$

$$\boxed{\sum_{f \in F} x_{f,d} \leq \text{MaxThroughput}_d \cdot w_d \quad \forall d \in D}$$

For Birmingham depot:

$$\boxed{\sum_{f \in F} x_{f,\text{Birmingham}} \leq (\text{MaxThroughput}_{\text{Birmingham}} + 20,000) \cdot e_b + \text{MaxThroughput}_{\text{Birmingham}} \cdot (1 - e_b)}$$

3. Customer demand constraints:

$$\boxed{\sum_{d \in D} y_{d,c} + \sum_{f \in F} z_{f,c} = \text{Demand}_c \quad \forall c \in C}$$

4. Depot opening limit:

$$\boxed{\sum_{d \in D_{\text{new}}} w_d + |D| - |D_{\text{close}}| - \sum_{d \in D_{\text{close}}} (1 - w_d) \leq 4}$$

Problem 9: Engineering factory

An engineering factory makes seven products (PROD_1 to PROD_7) on the following machines: four grinders, two vertical drills, three horizontal drills, one borer and one planer. Each product yields a certain contribution to profit. These quantities (in euros/unit) together with the production times (in hours/unit) required on each process are given below. A dash indicates that a product does not require a process.

	PROD_1	PROD_2	PROD_3	PROD_4	PROD_5	PROD_6	PROD_7
Contribution to profit (€/unit)	10	6	8	4	11	9	3
Grinding (hours/unit)	0.5	0.7	-	-	0.3	0.2	0.5
Vertical drilling (hours/unit)	0.1	0.2	-	0.3	-	0.6	-
Horizontal drilling (hours/unit)	0.2	-	0.8	-	-	-	0.6
Boring (hours/unit)	0.05	0.03	-	0.07	0.1	-	0.08
Planing (hours/unit)	-	-	0.01	-	0.05	-	0.05

In the present month (January) and the subsequent months, certain machines will be down for maintenance. These machines will be as follows:

Month	Machines Down for Maintenance
January	1 Grinder
February	2 Horizontal Drills
March	1 Borer
April	1 Vertical Drill
May	1 Grinder and 1 Vertical Drill
June	1 Planer and 1 Horizontal Drill

There are upper limits on how much each product can be sold in each month. These are given in the following table:

	PROD_1	PROD_2	PROD_3	PROD_4	PROD_5	PROD_6	PROD_7
January	500	1000	300	300	800	200	100
February	600	500	200	0	400	300	150
March	300	600	0	0	500	400	100
April	200	300	400	500	200	0	100
May	0	100	500	100	1000	300	0
June	500	500	100	300	1100	500	60

It is possible to store up to 100 units of each product at a time at a cost of 0.5 euros per unit per month. There are no stocks at present, but it is desired to have a stock of 50 of each type of product at the end of June. It may be assumed that each month consists of only 24 working days, each with two shifts of 8 hours each day.
(a) When and what should the factory make in order to maximise the total profit? Formulate the problem as an LP. (b) Instead of stipulating when each machine is down for maintenance, it is desired to find the best month for each machine to be down. Each machine must be down for maintenance in one month of the six apart from the grinding machines, only two of which need be down in any six months. Extend the model to allow it to make these extra decisions. Can you formulate it as an LP?
Solution:

(a) LP Formulation

Decision variables:

- $x_{i,m}$ = number of units produced of product i in month m (for all products i and months m) (of type real non negative)
- $s_{i,m}$ = number of units stored of product i at the end of month m (for all products i and months m) (of type real non negative)

Sets:

- Products: $P = \{\texttt{PROD_1}, \texttt{PROD_2}, \texttt{PROD_3}, \texttt{PROD_4}, \texttt{PROD_5}, \texttt{PROD_6}, \texttt{PROD_7}\}$
- Months: $M = \{\text{January}, \text{February}, \text{March}, \text{April}, \text{May}, \text{June}\}$
- Machines: $K = \{\text{Grinder}, \text{Vertical Drill}, \text{Horizontal Drill}, \text{Borer}, \text{Planer}\}$

Objective function: Maximize total profit minus storage costs:

$$\max \sum_{m \in M} \sum_{i \in P} \text{Profit}_i \cdot x_{i,m} - \sum_{m \in M} \sum_{i \in P} 0.5 \cdot s_{i,m}$$

Constraints:

1. Store maximum 100 units of each product:

$$\boxed{s_{i,m} \leq 100 \quad \forall i \in P, m \in M}$$

2. Machine capacity constraints:

$$\boxed{\sum_{i \in P} \text{Time}_{i,k} \cdot x_{i,m} \leq 24 * 8 * 2 * (|Machines_k| - |NonAvailable_{k,m}|)} \quad \forall k \in K, m \in M$$

3. Non negative production and storage:

$$\boxed{x_{i,m} \geq 0 \quad \forall i \in P, m \in M}$$

$$\boxed{s_{i,m} \geq 0 \quad \forall i \in P, m \in M}$$

4. Balancing production, storage and sales: Let $d_{i,m}$ be the demand (sales limit) for product i in month m.

$$\begin{aligned} &\text{For month } m = 1: && x_{i,1} = d_{i,1} + s_{i,1} \\ &\text{For month } 1 < m < 6: && x_{i,m_0-1} + s_{i,m_0-1} = d_{i,m_0} + s_{i,m_0} \\ &\text{For month } m = 6: && x_{i,6} + s_{i,5} = d_{i,6} + 50 \end{aligned}$$

Defining $s_{i,0} = 0$ and $s_{i,6} = 50$, we can express our constraints as:

$$\boxed{x_{i,m-1} + s_{i,m-1} = d_{i,m} + s_{i,m} \quad \forall i \in P, m \in M}$$

5. Production limits:

$$\boxed{x_{i,m} \leq \text{MaxProduction}_{i,m} \quad \forall i \in P, m \in M}$$

(b) MIP Formulation

Additional Decision variables:

- $y_{k,m} = 1$ if machine k is down for maintenance in month m, 0 otherwise (for all machines k and months m) (of type binary)

Additional Constraints:

1. Each machine must be down for maintenance in exactly one month:

$$\boxed{\sum_{m \in M} y_{k,m} = 1 \quad \forall k \in K, k \neq \text{Grinder}}$$

For Grinder machines (only two need to be down):

$$\boxed{\sum_{m \in M} y_{\text{Grinder},m} = 2}$$

2. Update machine availability based on maintenance schedule:

$$\boxed{\sum_{i \in P} \text{Time}_{i,k} \cdot x_{i,m} \leq 24 * 8 * 2 * (|Machines_k| - y_{k,m}) \quad \forall k \in K, m \in M}$$

Since we have introduced binary variables, the problem can no longer be formulated as an LP, and it must be treated as a MIP.

Problem 10: Mining company

A mining company is going to plan how to continue operating in a certain area for the next five years. There are four mines in this area, but it can operate at most three in any one year. Although a mine may not operate in a certain year, it is still necessary to keep it open, in the sense that royalties are payable, if it is to be operated in a future year. On the other hand, if a mine is not going to be worked again, it can be permanently closed down and no more royalties need be paid. The yearly royalties payable on each mine kept open are as follows:

Mine	Yearly Royalties
Mine 1	5 million €
Mine 2	4 million €
Mine 3	4 million €
Mine 4	5 million €

There is an upper limit on the amount of ore that can be extracted from each mine in a year. These upper limits are as follows:

Mine	Upper Limit
Mine 1	2×10^6 tons €
Mine 2	2.5×10^6 tons €
Mine 3	1.3×10^6 tons €
Mine 4	3×10^6 tons €

The ore from the different mines is of varying quality. The quality measurements for each mine are given in the following table:

Mine	Quality Measurement
Mine 1	1.0
Mine 2	0.7
Mine 3	1.5
Mine 4	0.5

Blending ores together results in an ore whose quality is the weighted average of the quality measurements of the ingredients. For example, if equal quantities of two ores were combined, the resultant ore would have a quality measurement half way between that of the ingredient ores. In each year, it is necessary to combine the total outputs from each mine to produce a blended ore of exactly some stipulated quality. For each year, these qualities are as follows:

Year	Required Quality
Year 1	0.9
Year 2	0.8
Year 3	1.2
Year 4	0.6
Year 5	1.0

The final blended ore sells for 10 e ton each year. Which mines should be operated each year and how much should they produce? Formulate the problem as a MIP.
Solution:

Decision variables:

- $x_{i,t}$ = amount of ore extracted from mine i in year t (for all mines i and years t) (of type real non negative)
- $y_{i,t} = 1$ if mine i is operated in year t, 0 otherwise (for all mines i and years t) (of type binary)
- $z_i = 1$ if mine i is kept open, 0 if permanently closed (for all mines i) (of type binary)

Sets:

- Mines: $I = \{\text{Mine 1}, \text{Mine 2}, \text{Mine 3}, \text{Mine 4}\}$
- Years: $T = \{1, 2, 3, 4, 5\}$

Objective function: Maximize total profit minus royalties:

$$\max \sum_{t \in T} \left(10 \cdot \sum_{i \in I} x_{i,t} - \sum_{i \in I} \text{Royalties}_i \cdot z_i \right)$$

Constraints:

1. Operate at most three mines in any year:

$$\boxed{\sum_{i \in I} y_{i,t} \leq 3 \quad \forall t \in T}$$

2. Upper limit on ore extraction from each mine:

$$\boxed{x_{i,t} \leq \text{UpperLimit}_i \cdot y_{i,t} \quad \forall i \in I, t \in T}$$

3. Linking operation and open status of mines:

$$\boxed{y_{i,t} \leq z_i \quad \forall i \in I, t \in T}$$

4. Quality blending constraint for each year:

$$\boxed{\sum_{i \in I} \text{Quality}_i \cdot x_{i,t} = \text{RequiredQuality}_t \cdot \sum_{i \in I} x_{i,t} \quad \forall t \in T}$$

Problem 11: Departments relocation

A large company wishes to move some of its departments out of London. There are bene ts to be derived from doing this (cheaper housing, government incentives, easier recruitment, etc.). Also, however, there will be greater costs of communication between departments. Where should each department be located so as to minimise overall yearly cost? The company comprises ve departments (A, B, C, D and E). The possible cities for relocation are Bristol and Brighton. Alternatively, a department may be kept in London. None of these cities (including London) may be the location for more than three of the departments. Bene ts to be derived from each relocation are given (in ke per year) as follows:

Department	Bristol	Brighton
A	10	10
B	15	20
C	10	15
D	20	15
E	5	15

Table 6: Relocation Benefits (in k€per year)

Communication costs are determined by C_{ik} and D_{jl}, where C_{ik} is the quantity of communication between departments i and k per year and D_{jl} is the cost per unit of communication between cities j and l.
Solution:
Decision variables:

- $x_{i,j} = 1$ if department i is located in city j, 0 otherwise (for all departments i and cities j) (of type binary)

Sets:

- Departments: $I = \{\text{A}, \text{B}, \text{C}, \text{D}, \text{E}\}$
- Cities: $J = \{\text{London}, \text{Bristol}, \text{Brighton}\}$

Objective function: Minimize total communication costs minus relocation benefits:

$$\min \sum_{i \in I} \sum_{k \in I, k \neq i} \sum_{j \in J} \sum_{l \in J} C_{ik} \cdot D_{jl} \cdot x_{i,j} \cdot x_{k,l} - \sum_{i \in I} \sum_{j \in J, j \neq \text{London}} \text{Benefit}_{i,j} \cdot x_{i,j}$$

The first term represents the total communication costs between departments based on their locations, while the second term accounts for the benefits derived from relocating departments out of London.
This is not linear due to the product of decision variables in the communication costs term. To linearize it, we can introduce additional variables and constraints, so we need an other decision variable:

- $w_{i,k,j,l} = 1$ if department i is located in city j and department k is located in city l, 0 otherwise (for all departments i, k and cities j, l) (of type binary)

Then we can rewrite the communication costs term as:

$$\sum_{i \in I} \sum_{k \in I, k \neq i} \sum_{j \in J} \sum_{l \in J} C_{ik} \cdot D_{jl} \cdot w_{i,k,j,l}$$

Constraints:

1. Each department must be located in exactly one city:

$$\boxed{\sum_{j \in J} x_{i,j} = 1 \quad \forall i \in I}$$

2. No more than three departments in any city:

$$\boxed{\sum_{i \in I} x_{i,j} \leq 3 \quad \forall j \in J}$$

3. Linking $w_{i,k,j,l}$ with $x_{i,j}$ and $x_{k,l}$:

$$\begin{aligned} w_{i,k,j,l} &\leq x_{i,j} \quad \forall i, k \in I, i \neq k, j, l \in J \\ w_{i,k,j,l} &\leq x_{k,l} \quad \forall i, k \in I, i \neq k, j, l \in J \\ w_{i,k,j,l} &\geq x_{i,j} + x_{k,l} - 1 \quad \forall i, k \in I, i \neq k, j, l \in J \end{aligned}$$

Problem 12: Power stations

A number of power stations are committed to meeting the following electricity load demands over a day:

Time Period	Electricity Load Demand
12p.m. to 6a.m.	15.000 MW
6a.m. to 9a.m.	30.000 MW
9a.m. to 3p.m.	25.000 MW
3p.m. to 6p.m.	40.000 MW
6p.m. to 12p.m.	27.000 MW

(The demands are for the whole indicated period, not per hour).
There are three types of generating units available: 12 of type 1, 10 of type 2, and 5 of type 3. Each generator has to work between a minimum and a maximum level. There is an hourly cost of running each generator at minimum level. In addition there is an extra hourly cost for each MW at which a unit is operated above minimum level. To start up a generator also involves a fixed cost. All this information is given in the table below:

Type	Minimum Level	Maximum Level	Cost per Hour at Minimum	Cost per Hour per MW Above Minimum	Startup Cost
Type 1	850 MW	2000 MW	1000 €	2 €	2000 €
Type 2	1250 MW	1750 MW	2600 €	1.3 €	1000 €
Type 3	1500 MW	4000 MW	3000 €	3 €	500 €

In addition to meeting the estimated load demands there must be sufficient generators working at any time to make it possible to meet an increase in load of up to 15%. This increase would have to be accomplished by adjusting the output of generators already operating within their permitted limits. Which generators should be working in which periods of the day to minimize total cost? Construct a MIP to answer this question.
Solution:
Decision variables:

- $x_{i,t}$ = amount of power generated by generator i in time period t (for all generators i and time periods t) (of type real non negative)
- $y_{i,t} = 1$ if generator i is operating in time period t, 0 otherwise (for all generators i and time periods t) (of type binary)
- $z_{i,t} = 1$ if generator i is started up in time period t, 0 otherwise (for all generators i and time periods t) (of type binary)

Sets:

- Generators: $I = \{\text{Type 1 (1-12)}, \text{Type 2 (1-10)}, \text{Type 3 (1-5)}\}$
- Time Periods: $T = \{\text{12p.m. to 6a.m.}, \text{6a.m. to 9a.m.}, \text{9a.m. to 3p.m.}, \text{3p.m. to 6p.m.}, \text{6p.m. to 12p.m.}\}$

Objective function: Minimize total cost:

$$\min \sum_{t \in T} \sum_{i \in I} (\text{CostMin}_i \cdot y_{i,t} + \text{CostPerMW}_i \cdot (x_{i,t} - \text{MinLevel}_i \cdot y_{i,t}) + \text{StartupCost}_i \cdot z_{i,t})$$

Constraints:

1. Meet electricity load demand with 15% increase:

$$\boxed{\sum_{i \in I} x_{i,t} \geq 1.15 \cdot \text{LoadDemand}_t \quad \forall t \in T}$$

2. Generator operating limits:

$$\boxed{\text{MinLevel}_i \cdot y_{i,t} \leq x_{i,t} \leq \text{MaxLevel}_i \cdot y_{i,t} \quad \forall i \in I, t \in T}$$

3. Linking startup variable: For the first time period:

$$\boxed{z_{i,1} \geq y_{i,1} \quad \forall i \in I}$$

For subsequent time periods:

$$\boxed{z_{i,t} \geq y_{i,t} - y_{i,t-1} \quad \forall i \in I, t \in T, t > 1}$$